SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
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# SYMBIOSIS INSTITUTE OF TECHNOLOGY

**Symbiosis International (Deemed University)**
(Established under Section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M.Sc., Ph.D.
(Awarded Padma Bhushan and Padma Shri by the President of India)

# A PROJECT REPORT

**ON**

## GuardianEye: Late Fusion of Image and Audio Poaching Indicators

Submitted in partial fulfillment of the requirements for the degree of
**Bachelor of Technology**
in
**Artificial Intelligence and Machine Learning**

**Subject: Big Data Analytics (BDA)**
**Course code: 0701260704**
**Academic Year: 2026–2027**

**Submitted by**


| Sr. No. | Name of the Student | PRN |
|---|---|---|
| 1 | Akhil Chivakula | 23070126009 |
| 2 | Dipti Kothari | 23070126040 |
| 3 | Khurinji Malar | 23070126059 |
| 4 | Sukhada Dhingra | 24070126505 |


Under the guidance of
**Dr. Archana Choudary**
Assistant Professor
**Department of Artificial Intelligence and Machine Learning**


Symbiosis Institute of Technology
Near Lupin Research Park, Gram Lavale, Taluka Mulshi
Pune – 412115, Maharashtra, India

# CERTIFICATE

This is to certify that the project work entitled “GuardianEye: Late Fusion of Image and Audio Poaching Indicators” is carried out by the students named below, of the Department of Artificial Intelligence and Machine Learning, Symbiosis Institute of Technology, Pune, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence and Machine Learning of Symbiosis International (Deemed University), Pune, during the academic year 2026–2027.

The work is submitted for the course Big Data Analytics (BDA), course code 0701260704, and was completed under the guidance of Dr. Archana Choudary.

| Sr. No. | Name of the Student | PRN |
|---|---|---|
| 1 | Akhil Chivakula | 23070126009 |
| 2 | Dipti Kothari | 23070126040 |
| 3 | Khurinji Malar | 23070126059 |
| 4 | Sukhada Dhingra | 24070126505 |

Place: Pune
Date: 26 October 2026

| Guide | Head of the Department |
|---|---|
| Dr. Archana Choudary | Dr. Sumit Kumar |
| Assistant Professor | Department of Artificial Intelligence and Machine Learning |
| Department of Artificial Intelligence and Machine Learning | Symbiosis Institute of Technology, Pune |
| Signature | Signature |

# DECLARATION

We hereby declare that the project report entitled “GuardianEye: Late Fusion of Image and Audio Poaching Indicators” is a record of work carried out by us at the Department of Artificial Intelligence and Machine Learning, Symbiosis Institute of Technology, Pune, a constituent of Symbiosis International (Deemed University), under the guidance of Dr. Archana Choudary, during the academic year 2026–2027.

This report is submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in Artificial Intelligence and Machine Learning, for the course Big Data Analytics (BDA), course code 0701260704.

We further declare that this report has not been submitted to any other university or institute for the award of any degree or diploma.

| Sr. No. | Name of the Student | PRN | Signature |
|---|---|---|---|
| 1 | Akhil Chivakula | 23070126009 | |
| 2 | Dipti Kothari | 23070126040 | |
| 3 | Khurinji Malar | 23070126059 | |
| 4 | Sukhada Dhingra | 24070126505 | |

Place: Pune
Date: 26 October 2026

Certified that the above statement made by the students is correct to the best of our knowledge and belief.

**Dr. Archana Choudary**
Guide
Assistant Professor
Department of Artificial Intelligence and Machine Learning
Symbiosis Institute of Technology, Pune

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Akhil Chivakula
Dipti Kothari
Khurinji Malar
Sukhada Dhingra

# ABSTRACT

GuardianEye is a two-modality review pipeline for wildlife evidence. Still frames from camera traps and from aerial thermal imagery pass through a detector trained separately on each domain. Short audio clips pass through a log-mel convolutional classifier. Video is sampled into frames and follows the image branch. The two branches meet only at decision time. A human box, a vehicle box, or a gunshot-like score is an indicator. One detection is not proof of poaching. No public collection used here carries a poacher label, and none aligns an image with the audio of the same moment.

The data design keeps that separation visible. Image counts stay split between aerial_thermal and camera_trap_rgb. Audio is split by site: Belize to Vietnam, Vietnam to Belize, and an in-domain control for each site. Empty frames and background clips remain in the prepared sets. Rare visual classes are oversampled. Audio training uses class-weighted loss. Fusion pairs are built from labels and are marked as a synthetic proxy protocol.

A local PySpark program, bda/guardianeye_bda_analysis.py, aggregates the prepared manifests, the site-stratified split files, the proxy pairs, and the SQLite audit log. On the corpus present in this working tree the job reports 22 images, 11 in each domain, of which 12 are empty; 29 audio windows, 14 from Belize and 15 from Vietnam, of which 20 are background; 84 split-assignment rows across four experiments; 16 synthetic proxy pairs; and one audit event, which was not alerted. Those figures describe metadata. Detection scores are written only by the evaluation scripts, and only when a checkpoint is present. The model directory contains no checkpoint.

*Keywords: late fusion, site stratification, class imbalance, synthetic proxy protocol, PySpark aggregation, audit log.*

# CONTENTS

Certificate
Declaration
Acknowledgement
Abstract
List of Tables
1. Introduction
2. Indicator Definition and Decision Policy
3. Data Pipeline
4. Image Branch
5. Audio Branch and Site Stratification
6. Synthetic Proxy Fusion
7. Serving and the Audit Log
8. Big Data Analytics Program
8.1 Design
8.2 Outputs
8.3 How to Run
9. Empirical Analysis of the Prepared Corpus
References

# LIST OF TABLES

Table 1. Image records by domain and primary class.
Table 2. Image records by domain and split.
Table 3. Audio windows by site and label.
Table 4. Audio variants and representations.
Table 5. Audio split sizes by experiment.
Table 6. Site composition of each audio split.
Table 7. Synthetic proxy pairs by scenario.
Table 8. One-hour window over the audit log.

# 1. Introduction

Patrol records arrive as still images, thermal frames, and short sound clips. The volume is uneven, the labels are coarse, and an image is seldom captured at the same instant as a sound. GuardianEye is the review pipeline built for that material. It searches for indicators in two modalities and fuses them late, at the decision, rather than inside a single joint network.

Video is not a third modality. Frames are sampled from it and submitted to the image branch. The image branch itself is two domains: aerial_thermal and camera_trap_rgb. A model trained on one is not scored as if it were the other. The audio branch distinguishes gunshot from background, and it is evaluated by site. Belize and Vietnam are held apart.

There is no labelled poacher class in the collections named by the configuration, and there is no time-aligned field incident that pairs an image with the audio of the same moment. The system therefore detects indicators. A human, a vehicle, or a gunshot-like sound is not, by itself, a finding that poaching occurred. One detection is not proof of poaching.

This report records the pipeline as it stands in the repository and the Big Data Analytics program that aggregates its prepared metadata. Configuration lives in configs/*.yaml. The seed is 27. Download entries in configs/data.yaml are the token TODO. The downloader skips those entries. The preparation path then materialises the generated sample tree, and the empirical counts in Chapter 9 are the counts of that prepared corpus. Evaluation scripts are the only component that writes detection scores, and they write a score only when the corresponding checkpoint is on disk. The models directory contains no checkpoint.

# 2. Indicator Definition and Decision Policy

## 2.1 Classes

Image classes are human, vehicle, animal, and empty. An empty image is a frame with no box. It is a negative example, and it stays in the training set. Box classes are human, vehicle, and animal. Visual indicators, the classes that can raise the image side of a fusion decision, are human and vehicle. An animal box is stored with the event. It does not enter the indicator score.

Audio labels are background and gunshot. The gunshot class is a gunshot-like impulse. Background covers ambient sound and also impulse-like distractors. The prepared manifest tags those distractors as variants, including thunder and snap. A spectrogram image, when one is supplied, is an audio representation. It is resized onto the log-mel grid. It is not a third modality, and the preparer does not invent a waveform for it.

## 2.2 Late fusion

The primary policy is rules mode, set by configs/fusion.yaml. A score equal to a threshold counts as strong. The image positive threshold and the audio positive threshold are both 0.50. When only one reliable modality is present, the threshold for that modality rises to 0.85, and the decision cannot be HIGH.

HIGH requires a human or vehicle detection and a gunshot score, both at or above threshold, with the detection timestamp inside the audio window expanded by 5.0 seconds. HIGH is the default alert. MEDIUM is issued when exactly one indicator is strong, or when both indicators are strong and fall outside that window. LOW is issued when neither indicator is strong. If a file was

uploaded but the corresponding checkpoint is missing, the evidence is marked unreliable. When neither modality is reliable, the decision is LOW and the reason code is unreliable_weights.

Three weight profiles change the numeric score. Balanced assigns 0.5 to each modality. Night assigns 0.3 to the image and 0.7 to the audio. Wind assigns 0.7 to the image and 0.3 to the audio. In rules mode those weights alter the score; the level still follows the indicator rules. Weighted mode maps the blended score onto HIGH at 0.75 and MEDIUM at 0.45, and it still refuses HIGH when a modality is missing. The alert list contains HIGH alone, so a MEDIUM or LOW decision is logged and is not raised as an alert.

# 3. Data Pipeline

## 3.1 Layout and configuration

Paths are relative to the repository root. The sample tree is data/sample, raw archives are data/raw, and prepared outputs are data/processed. The flag use_sample_if_raw_missing is true, so a missing or empty raw directory does not stop preparation. Image validation and test ratios are 0.15 each. The same ratios apply to audio. Splits are stratified by class inside one domain, or inside one site, and the ordering inside a class is the record identifier. The split is deterministic.

Accepted image layouts are an annotations.jsonl file, an annotations.csv file with columns filename, label, x1, y1, x2, y2, YOLO label files beside the images, or an empty directory of negatives. A blank label is a negative image. Strings pass through image.label_map. Person and pedestrian map to human. Car and truck map to vehicle. Named species map to animal. Empty, blank, and negative map to empty. An unknown label is skipped and logged.

## 3.2 Named collections

configs/data.yaml names five collections and sets every URL to TODO. A sha256 is checked only after a person replaces that token and supplies a digest. The image sources are BIRDSAI, the primary aerial-thermal collection; Caltech Camera Traps, including empty frames; and Snapshot Serengeti, a supporting camera-trap collection. Both camera-trap collections are domain camera_trap_rgb. The audio sources are the Belize tropical-forest gunshot collection (Katsis et al.), the primary site, and the Chu Mom Ray collection from Vietnam, the cross-site collection. Vietnam files may be waveforms or spectrogram images under gunshot and background folders. Doull et al. is not required by any script.

## 3.3 Thermal representation and windows

Aerial thermal frames are stored for an RGB backbone. The selected representation is replicate: the single thermal channel is copied into three identical channels. The alternative, pseudocolor, is a percentile-normalised iron-style false colour. The choice is image.thermal_representation in configs/data.yaml.

Audio preparation, driven by configs/data.yaml, resamples to 16 kHz and cuts windows of 3.0 seconds with a hop of 3.0 seconds. Each window becomes a log-mel matrix of 64 bins, using an FFT size of 1024, a hop length of 256 samples, and a band from 50 Hz to 8000 Hz. Inference and evaluation use the hop in configs/audio.yaml, which is 1.0 second, on the same 3.0 second window. The prepared manifest therefore holds non-overlapping windows. The service can score a clip on a finer hop.

### 3.4 What the aggregation reads

The Spark job reads four products: data/processed/images/manifest.jsonl, data/processed/audio/manifest.jsonl, the experiment files under data/processed/audio/splits/, data/processed/paired/events.jsonl, and, when it exists, data/events.sqlite. On this tree the image source tags are sample_birdsai and sample_camera_trap. Those tags identify the generated sample tree, coloured blocks and synthesised impulses, which preparation writes so the pipeline can be exercised. Chapter 9 reports that corpus. It does not report a field collection.

## 4. Image Branch

The image trainer runs one YOLOv8 model per domain. configs/image.yaml sets train_per_domain to true and names yolov8n.pt as the base weights. The full schedule is 50 epochs, image size 640, batch size 8, on CPU, with patience 15. A smoke schedule of one epoch at image size 320 exists so the script path can be checked. The service marks smoke weights as smoke. They are a path check.

Class imbalance on the image side is handled by oversampling, not by dropping empty frames. In a copy of the training split, rare box classes are duplicated until their instance counts meet the most common box class, with at most 20 copies of any one image. Empty label files are left in place. They are never the oversampling target. Augmentation on the full schedule covers HSV jitter, a rotation of 5 degrees, translation, scale, horizontal flip at probability 0.5, mosaic, and random erasing. Mosaic closes for the last 10 epochs.

Evaluation is per domain. guardianeye.eval.image_eval scores the test split of aerial_thermal and, separately, the test split of camera_trap_rgb. When weights exist it can write mAP50, mAP50-95, per-class precision and recall, small-object recall at the 1024-pixel-squared cutoff, and the empty-frame false-positive rate, into reports/image/<domain>.json. When the weight file is absent the same script writes a status record: evaluated is false, and the reason names the missing path. No number is inserted. No image checkpoint is present in this tree, so those JSON files are absent. The methodology is the script. The score is not.

## 5. Audio Branch and Site Stratification

The default audio model is a small convolutional network on one log-mel channel. Three blocks, each a 3×3 convolution, batch normalisation, and a ReLU, reduce the map. An adaptive pool yields a 4×4 grid at 64 channels, and a linear head emits two logits, background and gunshot. An optional head can sit on a frozen local embedding checkpoint in the PANNs style. The file must already be on disk, with an encoder module and an embedding width. The repository does not download that checkpoint and does not invent labels such as chainsaw.

Training uses class-weighted cross-entropy. Background clips remain in the loader. The weight of a class is the total count divided by the number of classes times that class count, so a rarer class receives a larger weight, and both classes stay in the tensor. On the training set the mel image may be shifted in time by up to 0.2, noised at standard deviation 0.01, and masked with a frequency width of 8 and a time width of 16. The full schedule is 20 epochs, batch size 16, learning rate 0.001, and weight decay 0.0001. The gunshot decision threshold is 0.5. The configured device is CPU.

Four experiments are defined. Validation is drawn only from the training site. In a cross-site experiment the other site is entirely test.

belize_to_vietnam trains and validates on Belize and tests on all of Vietnam. vietnam_to_belize reverses the sites. in_domain_belize and in_domain_vietnam each split one site into train, validation, and test. The in-domain rows are controls. They sit beside the transfer rows. They do not replace them. The service, when models/audio/best.pt is absent, looks for the checkpoint of the configured serve experiment, belize_to_vietnam.

Audio evaluation writes reports/audio/experiments.json. For each experiment that has a checkpoint, the file can hold precision, recall, F1, false-positive rate, and a precision-recall curve. A missing checkpoint produces a status record rather than a filled score. No audio checkpoint is present.

## 6. Synthetic Proxy Fusion

No public set used by this repository aligns an image with the audio of the same incident. scripts/build_paired_events.py therefore constructs pairs from labels. Every row carries protocol synthetic_proxy and a note that the pair is constructed and is not a field incident. The expected level on a row assumes a perfect indicator detector. It is a scenario label for the proxy protocol. It is not a model score.

Eight scenarios are applied in each image domain: human with gunshot, vehicle with gunshot, animal with background, empty with an impulse-noise background, vehicle with background, human with background, empty with gunshot, and a small human with gunshot. Impulse noise and the small-object case carry failure tags so a later analysis can list them. Animal with background is an expected LOW case, because an animal is not a visual indicator. A single strong indicator, such as a vehicle against background or an empty frame against a gunshot, is an expected MEDIUM case. Coincidence of a visual indicator and a gunshot is an expected HIGH case.

The output path is data/processed/paired/events.jsonl. Fusion evaluation, when both branches can be loaded, compares image-only, audio-only, and fused decisions on these pairs and writes reports/fusion/proxy.json. The caption required of that file is the same statement carried in configuration: synthetic proxy protocol, not field validation. The file is absent here because the checkpoints are absent.

## 7. Serving and the Audit Log

The service is a FastAPI application bound to port 8742. A Gradio dashboard is mounted at /dashboard. GET /health reports that the process is up. GET /model-info states, for each branch, whether a checkpoint loaded and whether it is a smoke run, a trained run, or missing. POST /events accepts a visual file, an audio file, or both, with a domain, a weight profile of balanced, night, or wind, and a decision mode of rules or weighted. GET /events returns the audit log, alerts and non-alerts together.

The audit store is SQLite at data/events.sqlite. The table events holds an event identifier, a timestamp, an integer alerted flag, and the JSON payload. Snapshots and audio copies are written under data/events/. A decision that does not alert is still inserted. The latency target in configs/serve.yaml is 5.0 seconds on CPU. A suite measurement of that latency is the responsibility of guardianeye.eval.latency, which writes reports/latency.json. That file is not

present. The latency quoted in Chapter 9 is the value stored on the single audit row and then averaged inside a one-hour Spark window. It describes one logged request.

# 8. Big Data Analytics Program

## 8.1 Design

bda/guardianeye_bda_analysis.py is the Big Data Analytics coursework artefact for the GuardianEye indicator pipeline. It performs a distributed-style aggregation of multimodal indicator metadata. It does not score a detector, and it is not a claim of field-validated poaching detection.

The engine is PySpark 4.0.1 on a local master with two threads. The Java runtime printed by the job is OpenJDK 21.0.12.1, dated 2026-08-18. Shuffle partitions are set to 2, the Spark user interface is disabled, and the Python random seed is 27, matching the pipeline seed. The process is a batch job on one machine. It does not require a remote cluster, and it does not launch training.

Each count is a map followed by a reduce. A manifest row is mapped to a key and the integer one. Keys are the image domain, the primary class, the split, the source tag, the audio site, the label, the variant, the representation, and, for the split files, the experiment together with the split, the site, and the label. reduceByKey sums the ones. The imbalance ratio inside a domain or a site is the largest class count divided by the smallest positive class count. Empty images and background clips participate in that ratio. They are not removed before the division.

The audit log is read with the standard SQLite library and then loaded into a Spark frame. A tumbling window of one hour groups the timestamps. Inside each bucket the job sums the alerted flag, counts the rows that were not alerted, and averages latency_ms. A second reduce groups the same rows by risk level, by reason code, and by domain.

## 8.2 Outputs

The job writes reports/bda/summary.json and reports/bda/aggregation.csv. The JSON note states that the file is an analytical summary of prepared indicator metadata, that it is not a set of model metrics, and that the fusion pairs use the synthetic proxy protocol. The CSV is the same tree flattened to section, key, and value.

## 8.3 How to Run

From the repository root, with the project virtual environment and with Java on the path, the command is:

```
.venv/bin/python bda/guardianeye_bda_analysis.py
```

The script is bda/guardianeye_bda_analysis.py. PySpark 4.0.1 is pinned in requirements.txt. The process prints a run log between the markers BDA RUN LOG and END BDA RUN LOG. Chapter 9 uses that log.

# 9. Empirical Analysis of the Prepared Corpus

The figures below are the figures printed by one execution of the Spark job on this working tree. The engine line was pyspark 4.0.1. The Java line was openjdk version "21.0.12.1" 2026-08-18.

The master was local[2]. The seed was 27. The full printed block is reproduced at the end of the chapter.

## 9.1 Images

The image manifest contains 22 records, 11 in aerial_thermal and 11 in camera_trap_rgb. The class counts are identical in the two domains: empty 6, human 2, animal 2, and vehicle 1. Twelve empty images are retained. Two images are marked small_object. The imbalance ratio, maximum count over minimum positive count, is 6.0 in each domain, which is the empty count divided by the vehicle count. The source tags are sample_birdsai for the 11 thermal frames and sample_camera_trap for the 11 RGB frames.

**Table 1. Image records by domain and primary class.**

| Domain | Human | Vehicle | Animal | Empty | Total | Imbalance |
|---|---|---|---|---|---|---|
| aerial_thermal | 2 | 1 | 2 | 6 | 11 | 6.0 |
| camera_trap_rgb | 2 | 1 | 2 | 6 | 11 | 6.0 |

The realised split, after per-class stratification at the configured ratios, is the same in both domains: 7 train, 3 validation, and 1 test. The two domains are stored separately. Nothing in the aggregation pools them into one image total for a detector score. The total of 22 is a record count.

**Table 2. Image records by domain and split.**

| Domain | Train | Validation | Test |
|---|---|---|---|
| aerial_thermal | 7 | 3 | 1 |
| camera_trap_rgb | 7 | 3 | 1 |

## 9.2 Audio

The audio manifest contains 29 windows. Belize contributes 14, of which 4 are gunshot and 10 are background. Vietnam contributes 15, of which 5 are gunshot and 10 are background. Twenty background clips are retained. The imbalance ratio is 2.5 at Belize (10 divided by 4) and 2.0 at Vietnam (10 divided by 5).

**Table 3. Audio windows by site and label.**

| Site | Gunshot | Background | Total | Imbalance |
|---|---|---|---|---|
| Belize | 4 | 10 | 14 | 2.5 |
| Vietnam | 5 | 10 | 15 | 2.0 |

Twenty-eight windows are stored as log-mel features. One window is a spectrogram image. Variants on the manifest are ambient 16, impulse_shot 8, thunder 2, snap 2, and spectrogram_image 1. Thunder and snap remain inside the background class. They are the impulse-noise cases the proxy protocol later tags.

**Table 4. Audio variants and representations.**

| Field | Value | Count |
|---|---|---|

| | | |
|---|---|---|
| representation | log_mel | 28 |
| representation | spectrogram_image | 1 |
| variant | ambient | 16 |
| variant | impulse_shot | 8 |
| variant | thunder | 2 |
| variant | snap | 2 |
| variant | spectrogram_image | 1 |

## 9.3 Site-stratified splits

The four experiment files together contain 84 assignment rows. That figure is a membership count. The supervised set is the 28 log-mel windows. Each of those windows is assigned in more than one experiment, so the rows sum to 84. The spectrogram image stays on the manifest and does not appear in the split files.

belize_to_vietnam has 11 training rows and 3 validation rows, all Belize, and 14 test rows, all Vietnam. vietnam_to_belize has 11 training rows and 3 validation rows, all Vietnam, and 14 test rows, all Belize. The cross-site test is the entire other site. in_domain_belize divides the 14 Belize windows into 8 train, 3 validation, and 3 test. in_domain_vietnam divides the 14 Vietnam log-mel windows the same way. Train and validation of a cross-site run never contain the test site.

**Table 5. Audio split sizes by experiment.**

| Experiment | Train | Validation | Test | Rows |
|---|---|---|---|---|
| belize_to_vietnam | 11 | 3 | 14 | 28 |
| vietnam_to_belize | 11 | 3 | 14 | 28 |
| in_domain_belize | 8 | 3 | 3 | 14 |
| in_domain_vietnam | 8 | 3 | 3 | 14 |

**Table 6. Site composition of each audio split.**

| Experiment | Train site | Validation site | Test site |
|---|---|---|---|
| belize_to_vietnam | Belize 11 | Belize 3 | Vietnam 14 |
| vietnam_to_belize | Vietnam 11 | Vietnam 3 | Belize 14 |
| in_domain_belize | Belize 8 | Belize 3 | Belize 3 |
| in_domain_vietnam | Vietnam 8 | Vietnam 3 | Vietnam 3 |

## 9.4 Synthetic proxy pairs

The pair file contains 16 rows. All 16 carry the protocol value synthetic_proxy. Eight rows are aerial_thermal and eight are camera_trap_rgb. Each of the eight scenarios occurs twice, once in each domain. The expected-level labels, which are scenario labels and not model scores, are HIGH 6, MEDIUM 6, and LOW 4.

**Table 7. Synthetic proxy pairs by scenario.**

| Scenario | Rows | Role in the proxy protocol |
|---|---|---|
| human_gunshot | 2 | Expected HIGH |
| vehicle_gunshot | 2 | Expected HIGH |
| small_human_gunshot | 2 | Expected HIGH, small-object tag |
| vehicle_background | 2 | Expected MEDIUM |
| human_background | 2 | Expected MEDIUM |
| empty_gunshot | 2 | Expected MEDIUM |
| animal_background | 2 | Expected LOW |
| empty_impulse | 2 | Expected LOW, impulse-noise tag |

## 9.5 Audit log

data/events.sqlite contains one event. The job reports alerted 0 and not_alerted 1. The risk level is LOW. The reason code is unreliable_weights. The domain on the payload is camera_trap_rgb. That reason is the fusion rule for a request in which neither modality has reliable weights, which is the state of this tree: no checkpoint is loaded. The one-hour tumbling window that contains the timestamp runs from 2026-10-26 10:00:00 to 2026-10-26 11:00:00. Inside that bucket the job reports 0 alerted, 1 not alerted, and a mean latency of 969.823 milliseconds. The figure is the mean of one logged request. It is not the output of the latency evaluation script.

**Table 8. One-hour window over the audit log.**

| Window start | Window end | Alerted | Not alerted | Mean latency (ms) |
|---|---|---|---|---|
| 2026-10-26 10:00:00 | 2026-10-26 11:00:00 | 0 | 1 | 969.823 |

## 9.6 Reading the counts

The empty class is the majority image class, and background is the majority audio class. Both were retained. The vehicle class is the minority image class, one frame in each domain, which produces the image imbalance of 6.0. Gunshot is the minority audio class at both sites. The cross-site experiments assign every supervised window of the held-out site to test, which is why each transfer test contains 14 rows while each in-domain test contains 3. The proxy file is balanced across the two image domains and across the eight scenarios. Its level counts are labels of those scenarios.

No precision, recall, F1, mAP, or false-positive rate is stated in this chapter, because the job does not compute them and the evaluation reports that would compute them are not on disk. The printed run log follows.

**Run log printed by bda/guardianeye_bda_analysis.py**

```
engine: pyspark 4.0.1
java: openjdk version "21.0.12.1" 2026-08-18
master: local[2]
seed: 27
image_records: 22
image_by_domain: {"aerial_thermal": 11, "camera_trap_rgb": 11}
image_by_domain_and_class: {"aerial_thermal": {"animal": 2, "empty": 6, "human": 2, "vehicle": 1}, "camera_trap_rgb": {"animal": 2, "empty": 6, "human": 2, "vehicle": 1}}
image_by_domain_and_split: {"aerial_thermal": {"test": 1, "train": 7, "val": 3}, "camera_trap_rgb": {"test": 1, "train": 7, "val": 3}}
image_by_source: {"sample_birdsai": 11, "sample_camera_trap": 11}
```

empty_images_retained: 12
small_object_images: 2
image_imbalance_ratio: {"aerial_thermal": 6.0, "camera_trap_rgb": 6.0}
audio_records: 29
audio_by_site: {"belize": 14, "vietnam": 15}
audio_by_site_and_label: {"belize": {"background": 10, "gunshot": 4}, "vietnam": {"background": 10, "gunshot": 5}}
audio_by_variant: {"ambient": 16, "impulse_shot": 8, "snap": 2, "spectrogram_image": 1, "thunder": 2}
audio_by_representation: {"log_mel": 28, "spectrogram_image": 1}
background_clips_retained: 20
audio_imbalance_ratio: {"belize": 2.5, "vietnam": 2.0}
audio_split_records: 84
audio_splits: {"belize_to_vietnam": {"test": 14, "train": 11, "val": 3}, "in_domain_belize": {"test": 3, "train": 8, "val": 3}, "in_domain_vietnam": {"test": 3, "train": 8, "val": 3}, "vietnam_to_belize": {"test": 14, "train": 11, "val": 3}}
audio_split_sites: {"belize_to_vietnam": {"test": {"vietnam": 14}, "train": {"belize": 11}, "val": {"belize": 3}}, "in_domain_belize": {"test": {"belize": 3}, "train": {"belize": 8}, "val": {"belize": 3}}, "in_domain_vietnam": {"test": {"vietnam": 3}, "train": {"vietnam": 8}, "val": {"vietnam": 3}}, "vietnam_to_belize": {"test": {"belize": 14}, "train": {"vietnam": 11}, "val": {"vietnam": 3}}}
proxy_records: 16
proxy_by_protocol: {"synthetic_proxy": 16}
proxy_by_scenario: {"animal_background": 2, "empty_gunshot": 2, "empty_impulse": 2, "human_background": 2, "human_gunshot": 2, "small_human_gunshot": 2, "vehicle_background": 2, "vehicle_gunshot": 2}
proxy_by_expected_level: {"HIGH": 6, "LOW": 4, "MEDIUM": 6}
proxy_by_domain: {"aerial_thermal": 8, "camera_trap_rgb": 8}
event_log_records: 1
event_log_alerted: 0
event_log_not_alerted: 1
event_log_by_risk_level: {"LOW": 1}
event_log_by_reason: {"unreliable_weights": 1}
event_log_by_domain: {"camera_trap_rgb": 1}
event_log_windows_1h: [{"alerted": 0, "end": "2026-10-26 11:00:00", "mean_latency_ms": 969.823, "not_alerted": 1, "start": "2026-10-26 10:00:00"}]

# References

[1] GuardianEye configuration files configs/data.yaml, configs/image.yaml, configs/audio.yaml, configs/fusion.yaml, and configs/serve.yaml. Repository working tree. Academic year 2026–2027. Download URLs in configs/data.yaml are recorded as TODO.

[2] Apache Spark 4.0.1 and PySpark 4.0.1. Local master local[2], as printed by bda/guardianeye_bda_analysis.py. OpenJDK 21.0.12.1.

[3] Ultralytics YOLOv8. Base weights named yolov8n.pt in configs/image.yaml. One training run is defined per image domain.

[4] Named source collections in configs/data.yaml, with URLs left as TODO: BIRDSAI (aerial thermal); Caltech Camera Traps, including empty frames; Snapshot Serengeti; the Belize tropical-forest gunshot collection (Katsis et al.); the Chu Mom Ray gunshot collection, Vietnam.

[5] SQLite audit table events, path data/events.sqlite, written by the FastAPI service on port 8742. Aggregated in reports/bda/summary.json.